

## Practical 2

**Aim:** Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

**Problem definition:**

You have recently joined DataAnalytics Inc. as a Hadoop Administrator. The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS).

As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

- Hadoop Installation & Configuration
- Access Hadoop Web UI.

1. Create directories in HDFS.
2. Upload files to HDFS.
3. List HDFS files and directories.
4. Display file contents.
5. Copy files within HDFS.
6. Move files within HDFS.
7. Rename files.
8. Download files from HDFS.
9. Delete files and directories.
10. Check HDFS report.
11. Verify block information.
12. Check Hadoop cluster health.
13. Monitor Hadoop processes.
14. Check storage utilization.

### Objectives:

- To install and configure Hadoop in a single-node environment on Ubuntu.
- To understand the architecture and working of HDFS.
- To perform basic HDFS file and directory operations.
- To manage files in HDFS using Hadoop commands.
- To monitor Hadoop services and cluster health.
- To analyze HDFS storage utilization and block information.
- To gain practical experience with Hadoop administration.

### Observation / Output:

1. Create directories in HDFS.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -mkdir /test1
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /
Found 2 items
drwxr-xr-x  - vboxuser supergroup          0 2026-07-15 17:08 /test
drwxr-xr-x  - vboxuser supergroup          0 2026-07-17 10:24 /test1
```

2. Upload files to HDFS.

```
vboxuser@Ubuntu:~$ echo "Hello World" > hello.txt
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -put hello.txt /
```

3. List HDFS files and directories.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /
Found 3 items
-rw-r--r--  1 vboxuser supergroup          12 2026-07-17 10:30 /hello.txt
drwxr-xr-x  - vboxuser supergroup          0 2026-07-15 17:08 /test
drwxr-xr-x  - vboxuser supergroup          0 2026-07-17 10:24 /test1
vboxuser@Ubuntu:~$
```

4. Display file contents.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -cat /hello.txt
Hello World
```

5. Copy files within HDFS.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -cp /hello.txt /hello_copy.txt
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /
Found 4 items
-rw-r--r--  1 vboxuser supergroup          12 2026-07-17 10:30 /hello.txt
-rw-r--r--  1 vboxuser supergroup          12 2026-07-17 10:32 /hello_copy.txt
drwxr-xr-x  - vboxuser supergroup          0 2026-07-15 17:08 /test
drwxr-xr-x  - vboxuser supergroup          0 2026-07-17 10:24 /test1
```

6. Move files within HDFS.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -mkdir /data
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -mv /hello_copy.txt /data/
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /data
Found 1 items
-rw-r--r--  1 vboxuser supergroup      12 2026-07-17 10:32 /data/hello_copy.txt
vboxuser@Ubuntu:~$
```

7. Rename files.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -mv /data/hello_copy.txt /data/sales_data.txt
```

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /data
Found 1 items
-rw-r--r--  1 vboxuser supergroup      12 2026-07-17 10:32 /data/sales_data.txt
vboxuser@Ubuntu:~$
```

8. Download files from HDFS.

```
vboxuser@Ubuntu:~$ mkdir download
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -get /data/sales_data.txt download/
vboxuser@Ubuntu:~$ ls download
sales_data.txt
vboxuser@Ubuntu:~$
```

9. Delete files and directories.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -rm /data/sales_data.txt
Deleted /data/sales_data.txt
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -rm /hello.txt
Deleted /hello.txt
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -rmdir /data
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfs -ls /
Found 2 items
drwxr-xr-x  - vboxuser supergroup      0 2026-07-15 17:08 /test
drwxr-xr-x  - vboxuser supergroup      0 2026-07-17 10:24 /test1
vboxuser@Ubuntu:~$
```

10. Check HDFS report.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfsadmin -report
Configured Capacity: 26238750720 (24.44 GB)
Present Capacity: 15568687058 (14.50 GB)
DFS Remaining: 15568625664 (14.50 GB)
DFS Used: 61394 (59.96 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
```

```
-----  
Live datanodes (1):  
  
Name: 127.0.0.1:9866 (localhost)  
Hostname: Ubuntu  
Decommission Status : Normal  
Configured Capacity: 26238750720 (24.44 GB)  
DFS Used: 61394 (59.96 KB)  
Non DFS Used: 9311268910 (8.67 GB)  
DFS Remaining: 15568625664 (14.50 GB)  
DFS Used%: 0.00%  
DFS Remaining%: 59.33%  
Configured Cache Capacity: 0 (0 B)  
Cache Used: 0 (0 B)  
Cache Remaining: 0 (0 B)  
Cache Used%: 100.00%  
Cache Remaining%: 0.00%  
Xceivers: 0  
Last contact: Fri Jul 17 10:40:28 UTC 2026  
Last Block Report: Fri Jul 17 10:22:24 UTC 2026  
Num of Blocks: 1
```

## 11. Verify block information.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs fsck /  
Connecting to namenode via http://localhost:9870/fsck?ugi=vboxuser&path=%2F  
FSCK started by vboxuser (auth:SIMPLE) from /127.0.0.1 for path / at Fri Jul 17 10:40:44 UTC 2026  
  
Status: HEALTHY  
Number of data-nodes: 1  
Number of racks: 1  
Total dirs: 3  
Total symlinks: 0  
  
Replicated Blocks:  
Total size: 13 B  
Total files: 1  
Total blocks (validated): 1 (avg. block size 13 B)  
Minimally replicated blocks: 1 (100.0 %)  
Over-replicated blocks: 0 (0.0 %)  
Under-replicated blocks: 0 (0.0 %)  
Mis-replicated blocks: 0 (0.0 %)  
Default replication factor: 1  
Average block replication: 1.0  
Missing blocks: 0  
Corrupt blocks: 0  
Missing replicas: 0 (0.0 %)  
Blocks queued for replication: 0  
  
Erasure Coded Block Groups:  
Total size: 0 B  
Total files: 0  
Total block groups (validated): 0  
Minimally erasure-coded block groups: 0  
Over-erasure-coded block groups: 0
```

```
Replicated Blocks:
Total size:      13 B
Total files:     1
Total blocks (validated):      1 (avg. block size 13 B)
Minimally replicated blocks:  1 (100.0 %)
Over-replicated blocks:       0 (0.0 %)
Under-replicated blocks:      0 (0.0 %)
Mis-replicated blocks:        0 (0.0 %)
Default replication factor:    1
Average block replication:     1.0
Missing blocks:                0
Corrupt blocks:                0
Missing replicas:              0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size:      0 B
Total files:     0
Total block groups (validated):      0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups:     0
Under-erasure-coded block groups:    0
Unsatisfactory placement block groups: 0
Average block group size:            0.0
Missing block groups:                0
Corrupt block groups:                0
Missing internal blocks:             0
Blocks queued for replication: 0
FSCK ended at Fri Jul 17 10:40:45 UTC 2026 in 7 milliseconds

The filesystem under path '/' is HEALTHY
```

## 12. Check Hadoop cluster health.

```
vboxuser@Ubuntu:~$ ~/hadoop/bin/hdfs dfsadmin -report
Configured Capacity: 26238750720 (24.44 GB)
Present Capacity: 15568646098 (14.50 GB)
DFS Remaining: 15568584704 (14.50 GB)
DFS Used: 61394 (59.96 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
```

13. Monitor Hadoop processes.

```
vboxuser@Ubuntu:~$ jps
5301 SecondaryNameNode
14025 Jps
5051 DataNode
4829 NameNode
```

14. Check storage utilization.

```
-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Ubuntu
Decommission Status : Normal
Configured Capacity: 26238750720 (24.44 GB)
DFS Used: 61394 (59.96 KB)
Non DFS Used: 9311412270 (8.67 GB)
DFS Remaining: 15568482304 (14.50 GB)
DFS Used%: 0.00%
DFS Remaining%: 59.33%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Jul 17 10:41:07 UTC 2026
Last Block Report: Fri Jul 17 10:22:24 UTC 2026
Num of Blocks: 1
```

**Result :**

Hadoop was successfully installed and configured in a single-node environment on Ubuntu Linux. HDFS services were started successfully, and all required file operations such as creating directories, uploading, downloading, moving, copying, renaming, and deleting files were performed successfully. Cluster health, storage utilization, and HDFS reports were verified, demonstrating the proper functioning of the Hadoop Distributed File System.

**Key Questions /Analysis:**

Q1. Why is Hadoop preferred for Big Data storage?

**Answer:**

Hadoop is preferred because it can store and process massive amounts of structured and unstructured data across multiple machines. It provides scalability, fault tolerance, high availability, and cost-effective storage using commodity hardware.

Q2. What is the role of NameNode and DataNode in HDFS?

**Answer:**

- **NameNode:** Maintains metadata of the file system, such as file names, permissions, and block locations.
- **DataNode:** Stores the actual data blocks and communicates with the NameNode to manage storage and retrieval.

The NameNode acts as the master, while DataNodes are the worker nodes.

Q.3 What is the difference between HDFS and traditional file systems?

| <b>HDFS</b>                          | <b>Traditional File System</b>   |
|--------------------------------------|----------------------------------|
| Distributed across multiple machines | Stored on a single machine       |
| Designed for Big Data                | Designed for normal file storage |
| Fault tolerant                       | Limited fault tolerance          |
| Stores very large files              | Best for small and medium files  |
| High throughput                      | Lower throughput for big data    |

Q.4 What are the advantages of distributed storage?

**Answer:**

- High scalability
- Fault tolerance through replication
- Better performance using parallel processing
- High availability
- Cost-effective storage
- Easy expansion by adding new nodes

**Supplementary Problems:**

## 1. Configure Hadoop pseudo-distributed mode.

```
vboxuser@Ubuntu:~/hadoop$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Ubuntu]
vboxuser@Ubuntu:~/hadoop$ jps
5516 NameNode
5932 SecondaryNameNode
6095 Jps
vboxuser@Ubuntu:~/hadoop$
```

## 2. Change HDFS replication factor and observe results.

```
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -ls /input
Found 1 items
-rw-r--r-- 1 vboxuser supergroup          13 2026-08-04 16:14 /input/sample.txt
vboxuser@Ubuntu:~/hadoop$ hdfs fsck /input/sample.txt -files -blocks
Connecting to namenode via http://localhost:9870/fsck?ugi=vboxuser&files=1&blocks=1&path=%2Finput%2Fsample.txt
FSCK started by vboxuser (auth:SIMPLE) from /127.0.0.1 for path /input/sample.txt at Tue Aug 04 16:14:39 UTC 2026

/input/sample.txt 13 bytes, replicated: replication=1, 1 block(s): OK
0. BP-795039297-127.0.1.1-1785859538118:blk_1073741825_1001 len=13 Live_repl=1

Status: HEALTHY
Number of data-nodes: 1
Number of racks:      1
Total dirs:           0
Total symlinks:       0

Replicated Blocks:
Total size:           13 B
Total files:          1
Total blocks (validated): 1 (avg. block size 13 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks:        0
Corrupt blocks:         0
Missing replicas:       0 (0.0 %)
Blocks queued for replication: 0

Total size:           13 B
Total files:          1
Total blocks (validated): 1 (avg. block size 13 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks:        0
Corrupt blocks:         0
Missing replicas:       0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size:           0 B
Total files:          0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Tue Aug 04 16:14:39 UTC 2026 in 22 milliseconds

The filesystem under path '/input/sample.txt' is HEALTHY
```

### 3. Upload a large dataset into HDFS.

```
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -put ~/Downloads/large_dataset.csv /datasets/  
put: `/home/vboxuser/Downloads/large_dataset.csv': No such file or directory  
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -mkdir /datasets  
mkdir: `/datasets': File exists  
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -put large_dataset.csv /datasets/  
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -ls /datasets  
Found 1 items  
-rw-r--r-- 1 vboxuser supergroup 337814 2026-08-04 16:21 /datasets/large_dataset.csv  
vboxuser@Ubuntu:~/hadoop$ hdfs dfs -du -h /datasets  
329.9 K 329.9 K /datasets/large_dataset.csv  
vboxuser@Ubuntu:~/hadoop$
```

### Tools /Technology used:

Windows 10 / Windows 11

OpenJDK 11 or above

Hadoop 3.x

Winutils.exe

Command Prompt / PowerShell

### Conclusion:

This practical provided hands-on experience in installing and configuring Hadoop on Ubuntu Linux. Various HDFS operations such as creating directories, uploading, downloading, copying, moving, renaming, and deleting files were performed successfully. Hadoop services, cluster health, storage utilization, and block information were also verified. The practical enhanced the understanding of Hadoop administration and HDFS management, which are essential for handling large-scale data in enterprise environments.